WORKED SOLUTION

STEP III 1991, Question 10

Pure • Roots of polynomials

STEP III 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q10

The key idea

The polynomial has only three non-zero coefficients, so almost all of the symmetric functions Σ_m vanish and the given identity (Newton's identity) collapses to a two-term recurrence. Beyond degree n the identity no longer applies, but something even simpler does: every root satisfies the equation itself, so multiplying it by α^k and summing over the roots links S_{n+k} to earlier power sums.

Solution

The power sums up to S_{n-1}

Comparing $x^n - qx^{n-1} + r$ with $\prod_i (x - \alpha_i) = x^n - \Sigma_1 x^{n-1} + \Sigma_2 x^{n-2} - \cdots + (-1)^n \Sigma_n$, we read off

$$\Sigma_1 = q, \qquad \Sigma_2 = \Sigma_3 = \cdots = \Sigma_{n-1} = 0, \qquad (-1)^n \Sigma_n = r.$$

Take $1 \le p \le n-1$ in the given identity. Every term containing $\Sigma_2, \ldots, \Sigma_p$ vanishes, because $p \le n-1$. (When $p = 1$ the identity is just $S_1 - \Sigma_1 = 0$.) What survives is

$$S_p - q\,S_{p-1} = 0 \quad (2 \le p \le n-1), \qquad S_1 = q.$$

So $S_1 = q$ and each power sum is q times the one before. By induction on p,

$$S_p = q^p \qquad \text{for } 1 \le p \le n-1,$$

as required. It is convenient to note $S_0 = n$ too, since each $\alpha_i^0 = 1$.

S_n, S_{n+1} and S_{n+2}

For $p = n$ the identity still applies, and now the last term is present:

$$S_n - q\,S_{n-1} + (-1)^n n\,\Sigma_n = 0.$$

Since $(-1)^n \Sigma_n = r$, this gives

$$S_n = q \cdot q^{n-1} - nr = q^n - nr.$$

Ask yourself The identity we were given stops at $p = n$. How do we get beyond it?

We use the equation itself. Each root satisfies $\alpha_i^n = q\alpha_i^{n-1} - r$. Multiplying by α_i^k (with $k \geq 0$) and summing over i gives

$$S_{n+k} = q\, S_{n+k-1} - r\, S_k \qquad (k \geq 0). \tag{$*$}$$

With $k = 0$ this gives $S_n = q^n - nr$ again, a reassuring cross-check. Now

$$S_{n+1} = q\, S_n - r\, S_1 = q^{n+1} - nqr - qr = q^{n+1} - (n+1)qr,$$
$$S_{n+2} = q\, S_{n+1} - r\, S_2 = q^{n+2} - (n+1)q^2 r - q^2 r = q^{n+2} - (n+2)q^2 r.$$

Here we used $S_1 = q$ and $S_2 = q^2$, which hold because $n \geq 5$ makes $2 \leq n - 1$.

The general formula

The pattern $S_n = q^n - nr$, $S_{n+1} = q^{n+1} - (n+1)qr$, $S_{n+2} = q^{n+2} - (n+2)q^2 r$ suggests

$$S_{n+m} = q^{n+m} - (n+m)\, q^m r \qquad \text{for } 0 \leq m \leq n-1.$$

We prove this by induction on m. The case $m = 0$ is $S_n = q^n - nr$, shown above.

Suppose the formula holds for some $m - 1$, where $1 \leq m \leq n - 1$. By $(*)$ with $k = m$,

$$\begin{aligned} S_{n+m} &= q\, S_{n+m-1} - r\, S_m \\ &= q(q^{n+m-1} - (n+m-1)q^{m-1} r) - r\, q^m \\ &= q^{n+m} - (n+m)\, q^m r, \end{aligned}$$

where we used $S_m = q^m$, valid because $1 \leq m \leq n - 1$. This is the formula for m. By induction, it holds for all $0 \leq m \leq n - 1$, that is, for $m < n$.

Ask yourself Where exactly does the condition $m < n$ come in?

Only in the step $S_m = q^m$. At $m = n$ we would need S_n, which is $q^n - nr$, not q^n, so the extra $-nr$ feeds in and the pattern breaks. In fact $(*)$ gives $S_{2n} = q^{2n} - 2nq^n r + nr^2$.

Checks

- With $r = 0$ the equation is $x^{n-1}(x - q) = 0$, with roots q and 0 ($n - 1$ times), so every $S_k = q^k$. All our formulae reduce to this.
- With $q = 0$ the roots are the nth roots of $-r$. Then $S_p = 0$ for $1 \leq p \leq n - 1$, and $S_n = n \cdot (-r) = -nr$, matching $q^n - nr$.

- Numerically, we found all roots for random q, r and $5 \leq n \leq 9$ and summed their powers. $S_p = q^p$ and the formula for S_{n+m} $(0 \leq m \leq n-1)$ agreed to about 10^{-12}, while at $m = n$ the formula visibly fails.

Common slips

- Forgetting the factor p on the last term of the identity, which gives $S_n = q^n - r$ instead of $q^n - nr$.
- Getting the sign of Σ_n wrong: the constant term is $(-1)^n \Sigma_n$, so $(-1)^n n\Sigma_n = nr$ whatever the parity of n.
- Applying the given identity with $p > n$, where it is not valid as stated.

Beyond STEP

These are Newton's identities, and $(*)$ shows that the power sums satisfy a linear recurrence whose characteristic polynomial is the original equation. In matrix language, S_k is the trace of A^k, where A is the companion matrix of the polynomial. The fact that any symmetric polynomial in the roots can be written in terms of the Σ_m is the fundamental theorem of symmetric polynomials, which leads on to Galois theory. Allenby, *Rings, Fields and Groups*, is a good place to see this algebra developed.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.